PUMaC Geometry Problems

Individual Finals, 2013–2025

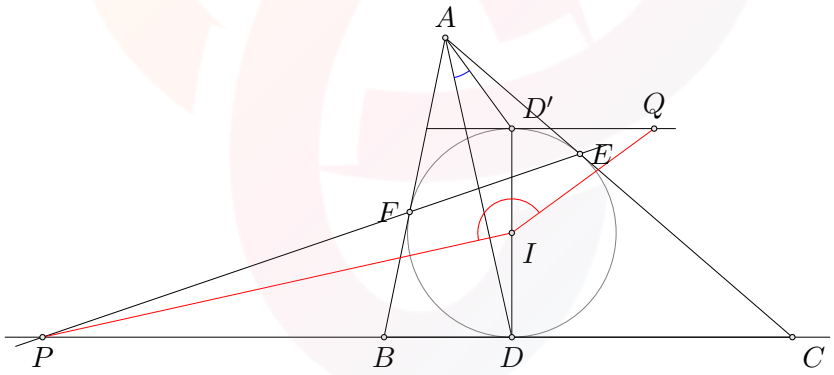
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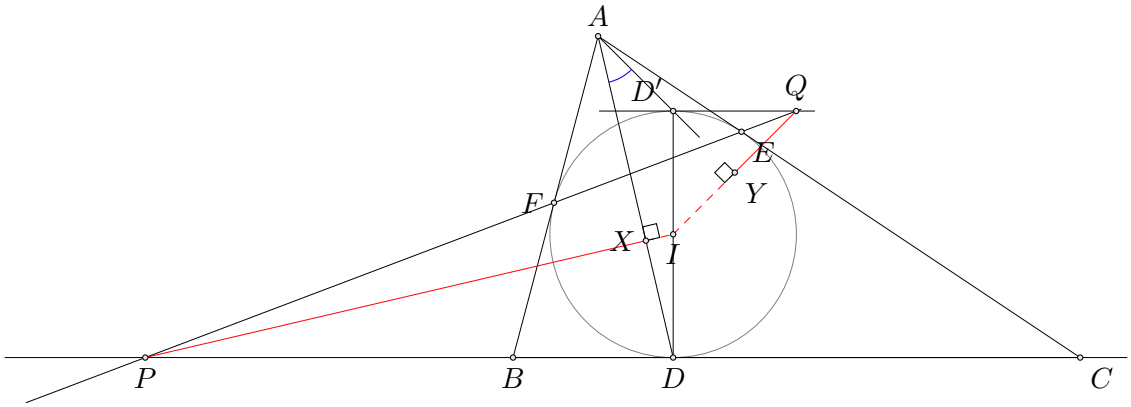
Problem 1

Source: PUMaC 2013 Individual Finals A, Problem 2.

Let γ be the incircle of $\triangle ABC$ (i.e. the circle inscribed in $\triangle ABC$) and I be the center of γ. Let D, E and F be the feet of the perpendiculars from I to BC, CA and AB respectively. Let D' be the point on γ such that DD' is a diameter of γ. Suppose the tangent to γ through D intersects the line EF at P. Suppose the tangent to γ through D' intersects the line EF at Q. Prove that $\angle PIQ + \angle DAD' = 180°$.



Solution.



All poles and polars are taken with respect to γ (*Beyond the Incenter*, Section 5.5). By Theorem 5.8, EF is the polar of A, while the tangents at D and D' are the polars of D and D'. Hence by *La Hire's Theorem* (Theorem 5.7), AD is the polar of P and AD' is the polar of Q.

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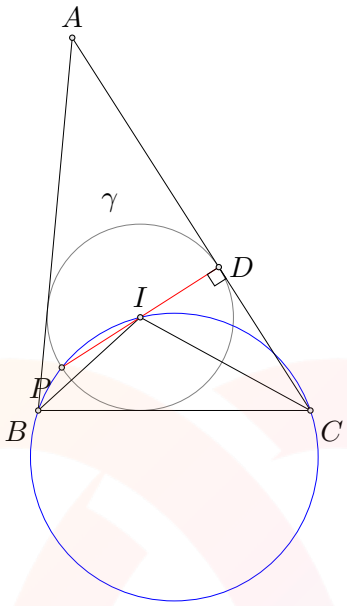
Let X, Y be the feet from I to AD, AD'. By definition of the polar, P lies on ray IX and Q on ray IY, so $\angle PIQ = \angle XIY$. Since $I \in DD'$ lies inside $\angle DAD'$ and X, Y lie on rays AD, AD' (each is the midpoint of a chord of γ), $AXIY$ is convex with right angles at X, Y. Thus $\angle PIQ = 180° - \angle DAD'$. ■

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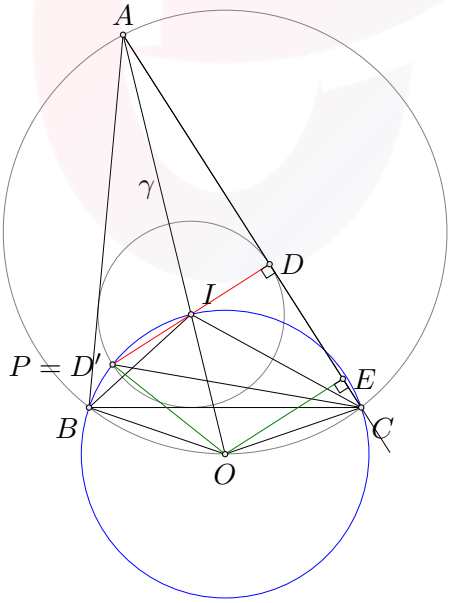
Problem 2

Source: PUMaC 2014 Individual Finals A, Problem 1.

Let γ be the incircle of $\triangle ABC$ (i.e. the circle inscribed in $\triangle ABC$) for which $AB + AC = 3BC$. Let the point where AC is tangent to γ be D. Let the incenter be I. Let the intersection of the circumcircle of $\triangle BCI$ with γ that is closer to B be P. Show that PID is colinear.



Solution.



Let O be the midpoint of arc BC of (ABC) not containing A. By the *Incenter–Excenter Lemma*, O is the centre of (BCI) with $OB = OC = OI$. *Ptolemy's Theorem* on $ABOC$ gives $AO \cdot BC = OC(AB + AC) = 3BC \cdot OC$, so $AO = 3OI$ and $AI = 2OI$.

Let D' be the antipode of D on γ, and let E be the foot from O to AC. Projecting A, I, O onto AC, we get $AE = \frac{3}{2}AD$, so $DE = \frac{1}{2}AD$, and O is at distance $\frac{3}{2}r$ from AC. Since D' is at distance $2r$ from AC on the same side,

$$OD'^2 = \left(\tfrac{r}{2}\right)^2 + DE^2 = \tfrac{1}{4}\left(r^2 + AD^2\right) = \tfrac{1}{4}AI^2 = OI^2.$$

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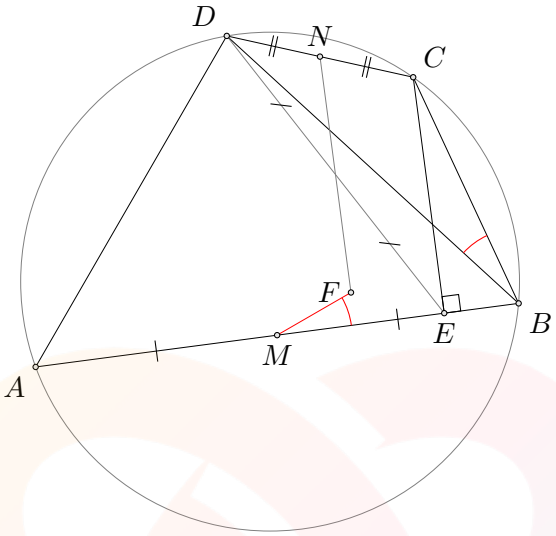
Hence $D' \in (BCI) \cap \gamma$. Both circles are centred on line AI, so their two intersections are reflections of each other in AI, and the one closer to B is the one on B's side of AI. As D lies on C's side, D' lies on B's side, so $P = D'$ and P, I, D are collinear. ■

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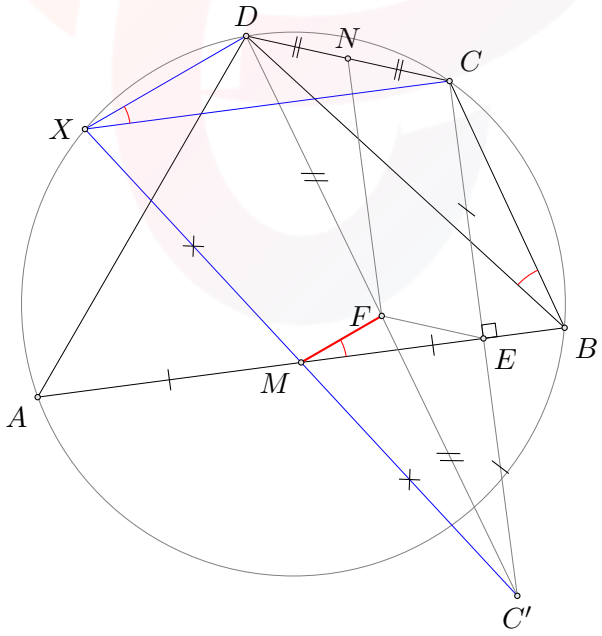
Problem 3

Source: PUMaC 2016 Individual Finals A, Problem 3.

On a cyclic quadrilateral $ABCD$, M is the midpoint of AB and N is the midpoint of CD. Let E be the projection of C onto AB and F the reflection of N about the midpoint of DE. If F is inside quadrilateral $ABCD$, show that $\angle BMF = \angle CBD$.



Solution.



Let C' be the reflection of C in AB and F' the midpoint of DC'. In $\triangle CDC'$, the midlines give $\overrightarrow{NF'} = \overrightarrow{CE}$ and $\overrightarrow{EF'} = \overrightarrow{CN} = \overrightarrow{ND}$, so $DNEF'$ is a parallelogram and $F' = F$. Thus F is the midpoint of DC'.

Let X be the reflection of C' in M, i.e. the reflection of C in the perpendicular bisector of AB. Then X lies on the circle with $XC \parallel AB$, and MF is a midline of $\triangle C'XD$, so $\overrightarrow{MF} = \frac{1}{2}\overrightarrow{XD}$.

As F is inside the convex quadrilateral $ABCD$, it lies on the same side of AB as C, so D is further from AB than C is. Hence D lies on the arc CX not containing A, B, so A, B, C, D, X lie on the

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circle in this order. Then $ABCX$ is an isosceles trapezoid, so $\overrightarrow{XC}$ points the same way as $\overrightarrow{MB}$, and B, X lie on the same arc CD. Therefore

$$\angle BMF = \angle CXD = \angle CBD.$$

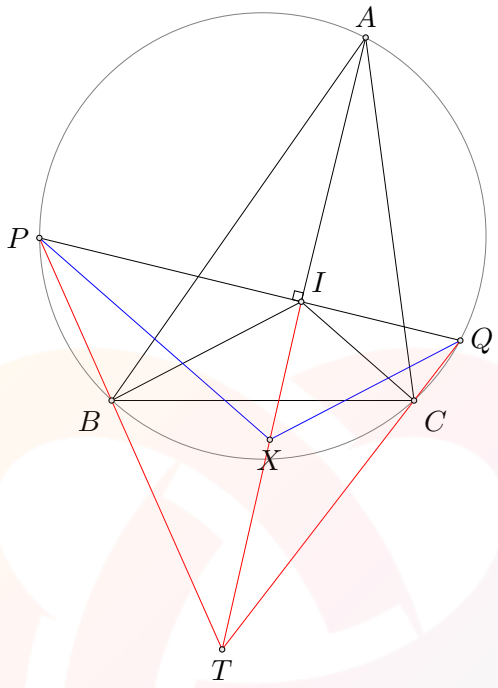
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Problem 4

Source: PUMaC 2017 Individual Finals A, Problem 3.

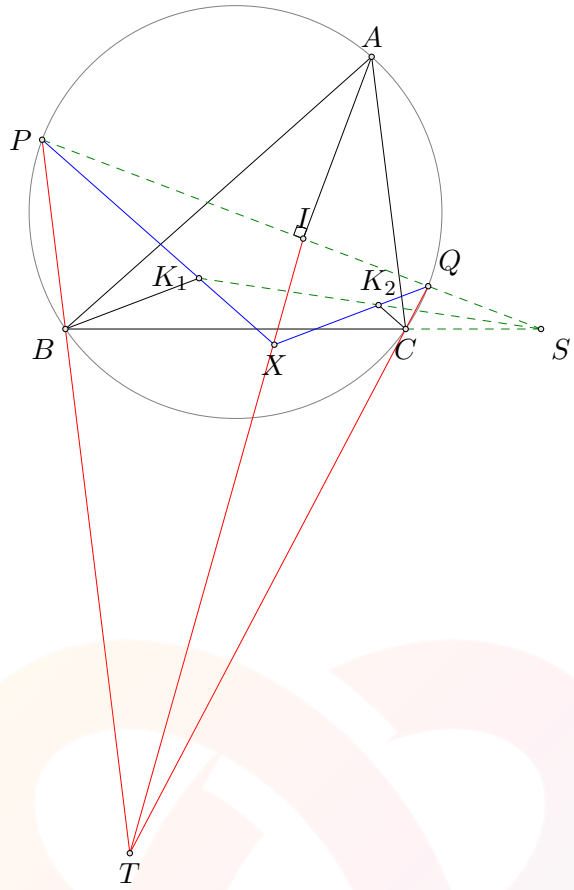
Triangle ABC has incenter I. The line through I perpendicular to AI meets the circumcircle of ABC at points P and Q, where P and B are on the same side of AI. Let X be the point such that $PX \parallel CI$ and $QX \parallel BI$. Show that PB, QC, and IX intersect at a common point.



Solution.

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Swapping B and C swaps P and Q, so we may assume $\angle B \le \angle C$. If $\angle B = \angle C$, the figure is symmetric in AI, so X and I lie on AI while PB and QC are reflections of each other in AI, and all three lines meet on AI (or are parallel to it). So let $\angle B < \angle C$, and write $\beta = \frac{1}{2}\angle B$, $\gamma = \frac{1}{2}\angle C$.

Let $T = PB \cap QC$, $K_1 = PX \cap BI$, $K_2 = QX \cap CI$ and $S = BC \cap PQ$. If $PB \parallel QC$, take T at infinity. Desargues' Theorem is projective, so the argument then shows that IX is parallel to both lines. In triangles TBC and XK_1K_2, the corresponding sides TB, XK_1 meet at P and TC, XK_2 meet at Q. So by *Desargues' Theorem*, the lines TX, $BK_1 = BI$ and $CK_2 = CI$ concur (that is, TX passes through I) as soon as K_1K_2 passes through S.

Locating $K_1K_2 \cap PQ$. Triangles IK_1P and QK_2I have parallel corresponding sides and $IP \neq IQ$ (shown below), so they are homothetic, and the centre is $Z = K_1K_2 \cap PQ$. The homothety sends $I \mapsto Q$ and $P \mapsto I$, so $ZQ/ZI = ZI/ZP$, i.e. $ZI^2 = ZP \cdot ZQ$. Hence Z lies outside segment PQ, and solving gives Z on ray IQ with

$$IZ = \frac{IP \cdot IQ}{IP - IQ},$$

provided $IP > IQ$, which we now check. The midpoint of PQ is the foot from O, and since $PQ \perp AI$ its distance from I is the distance from O to AI. Since AO is isogonal to the altitude from A, which makes angle $\gamma - \beta$ with AI on the side of C, the point O lies on the side of B with $\angle OAI = \gamma - \beta$. So the midpoint lies on ray IP and

$$IP - IQ = 2R\sin(\gamma - \beta), \qquad IP \cdot IQ = R^2 - OI^2 = 2Rr,$$

the latter by power of a point and *Euler's formula*. Thus $IZ = \dfrac{r}{\sin(\gamma - \beta)}$.

Locating S. Since $\angle AIB = 90^\circ + \gamma$ and $\angle AIP = 90^\circ$, we get $\angle BIP = \gamma$. Ray IB makes angle β with BC, so line PQ makes angle $\gamma - \beta$ with BC, with ray IP pointing away from BC. As I is at distance r from BC, the point S lies on ray IQ with $IS = \dfrac{r}{\sin(\gamma - \beta)}$.

Therefore $Z = S$, and PB, QC, IX concur at T. ■

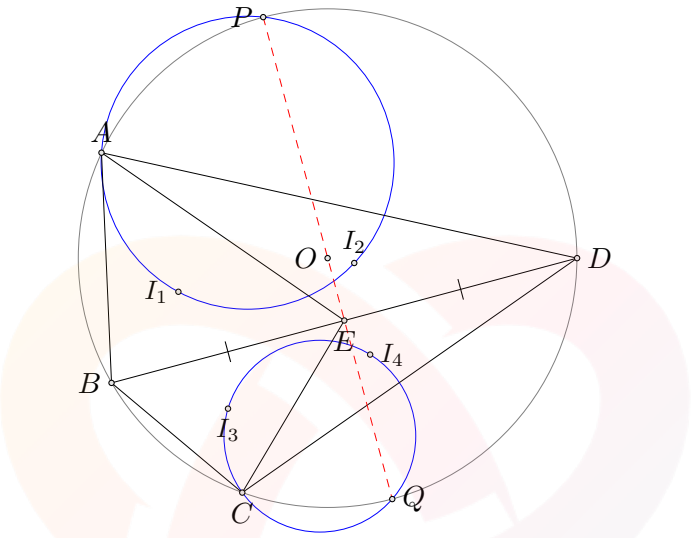
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Problem 5

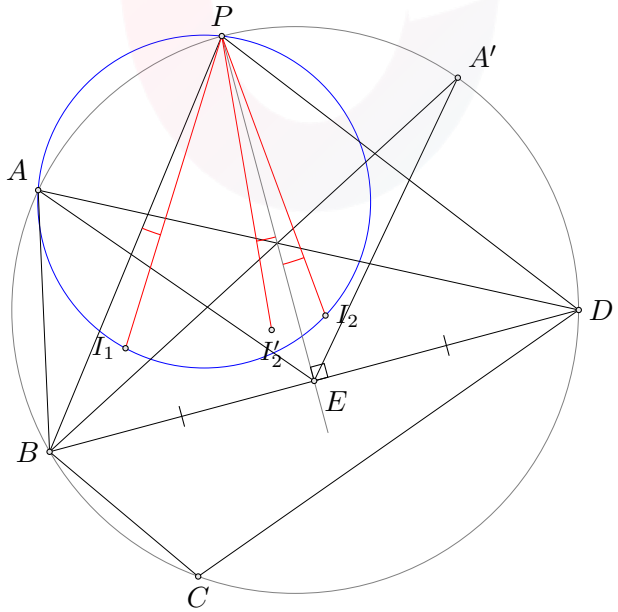
Source: PUMaC 2021 Individual Finals A, Problem 2.

Let $ABCD$ be a cyclic quadrilateral with circumcircle Γ, and let E be the midpoint of the diagonal BD. Let I_1, I_2, I_3, I_4 be the centers of the circles inscribed into triangles $\triangle ABE$, $\triangle ADE$, $\triangle BCE$, $\triangle CDE$, in that order. Prove that the circles AI_1I_2 and CI_3I_4 intersect Γ at diametrically opposite points.

Remark: For a circle C and points $X, Y \in C$, we say that X and Y are diametrically opposite if XY is a diameter of C.



Solution.



Let P and Q be the midpoints of arcs BD containing A and C respectively. Both lie on the perpendicular bisector of BD, which passes through E and the centre of Γ, so PQ is a diameter. It suffices to show $P \in (AI_1I_2)$, since $Q \in (CI_3I_4)$ follows in the same way.

Let σ be the reflection in line PE. It fixes Γ, P and E and swaps B and D. Let $A' = \sigma(A) \in \Gamma$ and $I_2' = \sigma(I_2)$, the incenter of $\triangle A'BE$.

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Claim. I_1 and I_2' are isogonal conjugates in $\triangle PBE$.

At B, the point P is the midpoint of arc AA' on arc BAD, so $\angle PBD = \frac{1}{2}(\angle ABD + \angle A'BD)$. Hence

$$\angle PBI_1 = \angle PBD - \tfrac{1}{2}\angle ABD = \tfrac{1}{2}\angle A'BD = \angle I_2'BE.$$

At E, since $PE \perp BD$,

$$\angle PEI_1 = 90^\circ - \tfrac{1}{2}\angle AEB = \tfrac{1}{2}\angle AED = \tfrac{1}{2}\angle A'EB = \angle I_2'EB.$$

Also $\angle EBI_1, \angle EBI_2' < \angle EBP$ and $\angle BEI_1, \angle BEI_2' < 90^\circ = \angle BEP$, so both points lie inside $\triangle PBE$, proving the claim. Therefore $\angle BPI_1 = \angle EPI_2' = \angle EPI_2$.

By symmetry I_2 lies inside $\triangle PDE$, so

$$\angle I_1PI_2 = \angle I_1PE + \angle BPI_1 = \angle BPE = \tfrac{1}{2}\angle BPD = \tfrac{1}{2}\angle BAD = \angle I_1AI_2.$$

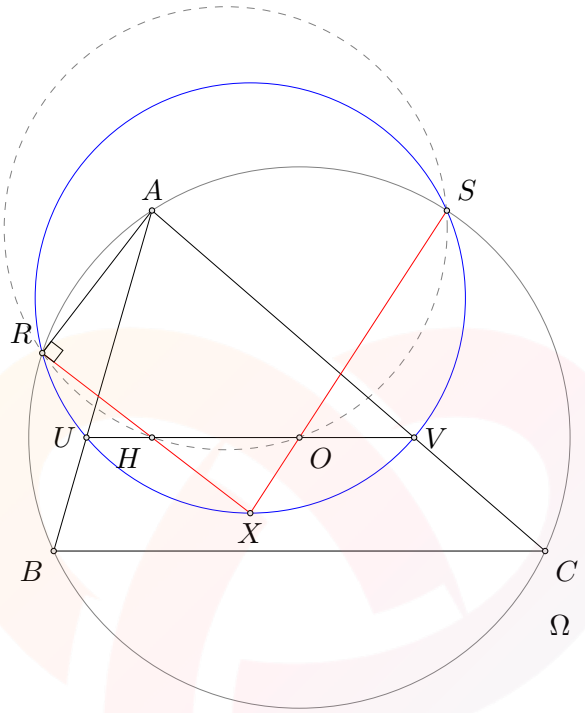
Finally, segment I_1I_2 crosses segment PE (as $I_1 \in \triangle PBE$, $I_2 \in \triangle PDE$) and segment AE (as $I_1 \in \triangle ABE$, $I_2 \in \triangle ADE$). So A and P lie on the same side of I_1I_2, and A, I_1, I_2, P are concyclic. Hence the two circles meet Γ again at P and Q, which are diametrically opposite. ■

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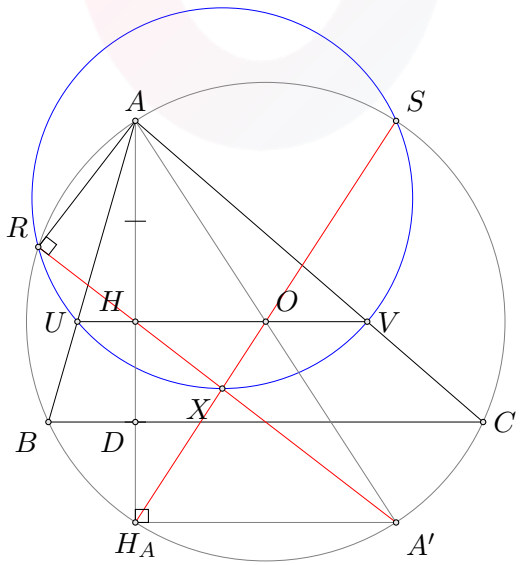
Problem 6

Source: PUMaC 2025 Individual Finals A, Problem 3.

Let ABC be an acute scalene triangle with orthocenter H, circumcenter O and circumcircle Ω. It is given that $OH \parallel BC$. Line OH meets side AB at U and side AC at V. Let R be on Ω such that $\angle ARH = 90°$. Let the circumcircle of triangle ROH meet Ω a second time at S. Show that lines SO and RH meet on the circumcircle of triangle RUV.



Solution.



Let D be the foot from A, let A' be the antipode of A, and let H_A be the second intersection of AD with Ω, so H_A is the reflection of H in BC. Since $\angle ARH = 90° = \angle ARA'$, the point R lies on line HA'. Also $\angle AH_AA' = 90°$, so $A'H_A \parallel BC \parallel OH$. As O is the midpoint of AA', OH is a midline of $\triangle AA'H_A$, so

$$AH = HH_A = 2HD, \qquad \overrightarrow{A'H_A} = -2\overrightarrow{HO}.$$

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Let $X = HA' \cap OH_A$ and let S' be the antipode of H_A. The relation above gives a homothety at X with ratio -2 sending $H \mapsto A'$ and $O \mapsto H_A$, so $XA' = 2XH$ and $XH_A = 2XO$. Power of a point at X in Ω gives

$$XR \cdot XA' = XS' \cdot XH_A \implies XR \cdot XH = XS' \cdot XO,$$

where R, H lie on the same side of X, and so do O, S'. Hence $S' \in (ROH)$. Also $S' \neq R$, as otherwise R would lie on both HA' and OH_A, forcing $R = X$, which lies inside Ω. So $S' = S$, and SO meets RH at X.

It remains to show $X \in (RUV)$. From the homothety, $HX = \frac{1}{3}HA'$, and power of a point at H gives $HR \cdot HA' = HA \cdot HH_A = AH^2$, so

$$HR \cdot HX = \tfrac{1}{3}AH^2.$$

Since $AH = \frac{2}{3}AD$, the homothety at A with ratio $\frac{2}{3}$ sends B, C, D to U, V, H. Using $DB \cdot DC = DA \cdot DH_A = AD \cdot HD$,

$$HU \cdot HV = \tfrac{4}{9}\, DB \cdot DC = \tfrac{4}{9} \cdot \tfrac{3}{2}AH \cdot \tfrac{1}{2}AH = \tfrac{1}{3}AH^2.$$

As H lies between R and X and between U and V, the converse of power of a point shows that R, U, V, X are concyclic. ■

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